# When Did the Lamb Die?

*An Astronomical Investigation*
*into the Year of the Crucifixion*

• • •

Paul & Pam • NobleMind.Study

*A study from Scripture and astronomical data. No commentaries, no denominational positions.*

*All Scripture quotations from the New American Standard Bible (NASB).*

## The Question

In the companion study, The Last Week of the Lamb, we established from the Gospel text that the crucifixion of Jesus occurred on a Wednesday — Nisan 14 on the Hebrew calendar. This conclusion was reached through multiple converging lines of textual evidence: Jesus' own prediction of three days and three nights (Matthew 12:40), John's identification of the following Sabbath as "a high day" (John 19:31), and the two-Sabbath spice sequence (Luke 23:56 and Mark 16:1). No other day of the week satisfies all of the textual constraints simultaneously.

If this is correct — if the crucifixion occurred on a Wednesday, Nisan 14 — then a question naturally follows: can we determine which year it happened?

The Hebrew calendar is lunar. Each month begins with the sighting of the new crescent moon. Nisan 14 is always fourteen days after the start of Nisan. If we can determine when the new moon of Nisan occurred in the years surrounding the crucifixion, we can calculate what day of the week Nisan 14 fell on in each year. And if only one year in the plausible range produces a Wednesday, we may have our answer.

This investigation uses astronomical data — not tradition, not commentary, not denominational positions. The astronomy is verifiable. The text is verifiable. What follows is an honest examination of what they tell us when brought together.

# The Constraints from the Text

Before looking at the astronomical data, we need to identify the boundaries the text gives us. Any proposed year must satisfy all of these constraints simultaneously.

## Constraint 1: Pontius Pilate Must Be in Office

All four Gospels identify Pontius Pilate as the Roman governor who sentenced Jesus to death. The Roman historian Tacitus confirms this (Annals, XV.44). Pilate governed Judaea from AD 26 to AD 36. The crucifixion must fall within this window.

## Constraint 2: The Fifteenth Year of Tiberius

> **Luke 3:1–3** — *"Now in the fifteenth year of the reign of Tiberius Caesar, when Pontius Pilate was governor of Judea… the word of God came to John, the son of Zacharias, in the wilderness."*

Luke anchors the beginning of John the Baptist's ministry to a specific year of Tiberius's reign. Tiberius's sole reign began in August AD 14. By standard Roman reckoning, his fifteenth year runs from approximately January AD 29 to January AD 30. Some scholars have proposed an earlier co-regency reckoning that could push this back to AD 26–27, but the straightforward reading of Luke places John's ministry beginning around AD 28–29.

Jesus was baptized by John after John began preaching. His public ministry followed. This gives us our earliest possible starting point for Jesus' ministry.

## Constraint 3: Three Passovers in John's Gospel

John records at least three distinct Passovers during Jesus' public ministry: the first near the beginning of His ministry (John 2:13), a second during His Galilean ministry (John 6:4), and the final Passover at which He was crucified (John 11:55). This means His ministry spanned at least two full years — and likely closer to three, since many scholars believe John omitted a fourth Passover, giving a ministry of approximately three and a half years.

If the ministry began around AD 29 and lasted through three Passovers, the crucifixion Passover falls around AD 31. If it lasted through four Passovers, it could extend to AD 32 or 33.

## Constraint 4: Nisan 14 Must Fall on a Wednesday

As established in the companion studies, the text requires a Wednesday crucifixion. Any proposed year must have Nisan 14 falling on a Wednesday.

## Constraint 5: Paul's Conversion

The apostle Paul's conversion is generally dated to approximately AD 33–34, based on working backward from his appearance before the proconsul Gallio in Corinth (Acts 18:12–17), which is well-dated to AD 51–52, and tracing his travels and timeline references in Galatians and Acts. The crucifixion must precede Paul's conversion. This provides a practical upper boundary of approximately AD 33–34.

## Summary of Constraints

The crucifixion must fall within a year that satisfies all of the following: Pilate is in office (AD 26–36), John the Baptist has already begun preaching (no earlier than AD 28–29), at least three Passovers have occurred during Jesus' ministry, Nisan 14 falls on a Wednesday, and the date is early enough to precede Paul's conversion. The practical window is approximately AD 29 to AD 34.

## The Astronomical Data

Astronomers can calculate the precise times of new moon conjunctions thousands of years into the past. The math is well-established and has been verified against ancient Babylonian and Chinese eclipse records with probable errors of only a few minutes, even for dates 2,000 years ago.

However, the first-century Jewish calendar was not based on the calculated astronomical new moon (the conjunction, which is invisible). It was based on the observed first visible crescent of the new moon — the thin sliver of light that appears after sunset, typically one to two days after the conjunction. Each month began on the evening when this crescent was first spotted by observers in Jerusalem.

This distinction is critical. We can calculate when the astronomical new moon occurred with extraordinary precision. But we cannot know with certainty when human eyes in Jerusalem first saw the crescent. Cloud cover, atmospheric haze, dust storms, or simple variation in observer skill could delay the sighting by a day. A one-day delay in sighting shifts the start of the month by one day — and therefore shifts Nisan 14 by one day.

The most rigorous modern work on this problem was done by Colin Humphreys and W. Graeme Waddington of Oxford University, who published their calculations in Nature (1983) and the Journal of the American Scientific Affiliation (1985). They computed the visibility of the lunar crescent as a function of time after sunset for each relevant month, taking into account the moon's position, semi-diameter, sky brightness, and visual contrast threshold for observers at Jerusalem's latitude. Their work has been widely cited and substantially confirmed by subsequent researchers.

Additional calculations come from Sir Isaac Newton (published posthumously in 1733), Jack Finegan's Handbook of Biblical Chronology, and various other researchers. While these sources use slightly different assumptions about crescent visibility and about the insertion of leap months, they converge on a consistent picture.

### The Day of the Week for Nisan 14: AD 27–34

The following table compiles results from multiple independent sources. Where sources disagree on the exact Julian date (due to differing assumptions about crescent visibility or leap months), the range is noted. The day of the week is the critical column for our investigation.

| Year | Nisan 14 (Julian Date) | Day of Week | Wednesday? |
|---|---|---|---|
| AD 27 | Mar 28 or Apr 11 | Friday | No |
| AD 28 | Apr 14 or Apr 28 | **Wednesday** | **Yes** |
| AD 29 | Apr 4–5 or Apr 18 | Monday | No |
| AD 30 | Apr 7 | Friday | No |
| AD 31 | Mar 27–28 or Apr 11–25 | **Tues/Wed** | **Probable** |

| | | | |
|---|---|---|---|
| AD 32 | Apr 14 | Monday | No |
| AD 33 | Apr 3 | Friday | No |
| AD 34 | Mar 24 or Apr 22–23 | Wed/Fri | Possible |

Three years produce a Wednesday for Nisan 14: AD 28, AD 31, and possibly AD 34.

# Applying the Constraints

## AD 28: Too Early

If John the Baptist began preaching in the fifteenth year of Tiberius (AD 28–29), an AD 28 crucifixion leaves virtually no time for John's ministry, Jesus' baptism, or a public ministry spanning three Passovers. John's Gospel alone records events across at least two and a half years of ministry. AD 28 does not provide enough time. It can be set aside.

## AD 34: Too Late

AD 34 conflicts with the probable date of Paul's conversion (approximately AD 33–34). It also requires assuming that an additional leap month was inserted due to unusually severe weather — a possibility, but one with no supporting historical evidence for that specific year. Furthermore, AD 34 is only a Wednesday under certain calculation methods; others place it on a Friday. The evidence is thin and speculative. It can be set aside.

## AD 31: The Strongest Candidate

AD 31 is the only year in the AD 29–33 window where Nisan 14 plausibly falls on a Wednesday. Multiple independent calculations point to this year:

Newton (1733), applying the standard crescent visibility rule and the Jewish postponement practice, calculated Nisan 14 in AD 31 as Wednesday, March 28. Modern researchers using different methods have placed Nisan 14 in AD 31 on various dates in late March or April, with the day of the week falling on either Tuesday or Wednesday depending on the assumptions used.

The Tuesday-versus-Wednesday variation reflects the observational uncertainty in detecting the new crescent moon — a one-day difference in sighting shifts the entire month by one day. Multiple calculations cluster around Wednesday; none of the major sources place it on a day that rules out Wednesday entirely.

Now check it against the textual constraints:

Pilate is in office: Yes. AD 31 falls within his governorship (AD 26–36).

John the Baptist has begun preaching: Yes. If John began in AD 28–29, he has been preaching for one to two years by AD 31.

Three Passovers during Jesus' ministry: If Jesus' ministry began around AD 29, the three Passovers recorded in John would fall in AD 29, AD 30, and AD 31. The crucifixion occurs at the third. This gives a ministry of approximately two and a half years, which is consistent with the minimum required by John's Gospel.

Early enough for Paul’s conversion: Yes. A crucifixion in AD 31 allows two to three years before Paul’s probable conversion in AD 33–34, during which the early church in Jerusalem would grow and the persecution that triggered Paul’s involvement would develop.

Every textual constraint is satisfied. No other year in the range does this while also producing a Wednesday.

# The Honest Limitations

Before stating any conclusion, we must acknowledge what we do not and cannot know.

## The Observational Calendar

The first-century Jewish calendar was determined by direct observation, not by mathematical calculation. Each month began when the thin crescent of the new moon was first spotted by observers in Jerusalem, typically one to two days after the astronomical conjunction. We can calculate the conjunction with extraordinary precision. We cannot know exactly when human eyes saw the crescent. Cloud cover, atmospheric transparency, dust from a khamsin wind, or even a one-day dispute among the observers could shift the start of the month — and therefore Nisan 14 — by one day.

This introduces an irreducible uncertainty of plus or minus one day. We can say that Nisan 14 in AD 31 very probably fell on a Wednesday. We cannot say it with absolute certainty.

## Leap Months

The Hebrew calendar periodically inserted a thirteenth month (a leap month before Nisan) to keep the lunar calendar aligned with the agricultural seasons. In the first century, this decision was made annually by the Sanhedrin based on whether the barley crop would be ripe enough by the time of the Feast of Firstfruits (Nisan 16) and whether Passover would fall after the vernal equinox. We have no historical records of when leap months were actually proclaimed in the AD 26–36 period. If a leap month was inserted in a year when modern calculations do not assume one (or vice versa), the date of Nisan 14 could shift by approximately thirty days, potentially changing both the Julian date and the day of the week.

For AD 31, the astronomical data does not suggest an obvious need for a leap month under normal conditions. But “normal conditions” is an assumption. Unusually late barley growth, heavy rains, or other agricultural delays could have triggered an insertion that we cannot detect from this distance.

## Postponement Rules

The modern Jewish calendar (formalized by Hillel II around AD 359) includes specific rules for postponing the start of certain months to avoid calendar conflicts such as two Sabbaths falling consecutively. Whether these postponement rules were in use in the first century is debated among scholars. Newton applied one such rule (called Adu) and arrived at his Wednesday, March 28 date for AD 31. Without that rule, his calculation gives Tuesday, March 27. The difference is one day — which falls within the observational uncertainty already noted. But it is worth stating plainly that different assumptions about postponement rules produce different results.

## What This Means

None of these limitations invalidate the investigation. They define its boundaries. Astronomical data is precise. The text is clear. But the bridge between them — the actual observational practices of first-century Jerusalem — carries a degree of uncertainty that cannot be eliminated with the information currently available.

Honesty requires us to say so.

## A Note on Prior Scholarship

Nearly all of the astronomical work done on dating the crucifixion — from Newton in 1733 to Humphreys and Waddington in 1983 to researchers publishing today — has been conducted under a single assumption: that the crucifixion occurred on a Friday. This assumption comes from the tradition of "Good Friday," which is itself based on the further assumption that "the day of preparation before the Sabbath" must mean Friday (the day before the weekly Sabbath).

As we have shown in the companion studies, this assumption does not hold when tested against the full weight of the textual evidence. John 19:31 explicitly identifies the Sabbath following the crucifixion as "a high day" — not the weekly Sabbath but the feast-day Sabbath of Nisan 15. The spice sequence in Luke 23:56 and Mark 16:1 requires two Sabbaths with a working day between them. And three literal days and three literal nights cannot fit between Friday afternoon and Sunday dawn.

The consequence is significant. When Humphreys and Waddington identified AD 30 and AD 33 as the only two years where Nisan 14 fell on a Friday, they were solving for the correct day of the month (Nisan 14) but the wrong day of the week (Friday). Their astronomical calculations are sound. Their Nisan 14 identification is consistent with the text. Their Friday requirement is not.

When the same astronomical data is examined for years where Nisan 14 falls on a Wednesday instead of a Friday, the field changes entirely. AD 30 and AD 33 are eliminated. AD 31 emerges as the strongest candidate.

This is not a criticism of these researchers' astronomical methods, which are excellent. It is a correction of the inherited assumption they were working from. The tradition was old. The text is older.

# Conclusion

## What We Can Say

The textual evidence from all four Gospels establishes that Jesus was crucified on Nisan 14, a Wednesday, with burial completed before sundown. This is a necessary inference from the convergence of Matthew 12:40, John 19:31, Leviticus 23:5–7, and the Luke/Mark spice sequence.

The astronomical data shows that within the plausible window for the crucifixion (approximately AD 29–34), only one year produces a Wednesday for Nisan 14: AD 31. This year satisfies every textual constraint — Pilate's governorship, the timing of John the Baptist's ministry, the three Passovers in John's Gospel, and the chronology of Paul's conversion.

AD 31 is the strongest candidate for the year of the crucifixion.

## What We Cannot Say

We cannot say with absolute certainty that Nisan 14 fell on a Wednesday in AD 31. The observational nature of the first-century Jewish calendar introduces an irreducible uncertainty of plus or minus one day. The astronomical calculations cluster around Wednesday, but the actual date depended on when human observers in Jerusalem first spotted the new crescent moon — and that information is not recoverable.

We also cannot rule out the possibility that unusual agricultural conditions triggered a leap month insertion that would shift the calendar by approximately thirty days, although no historical evidence suggests this occurred in AD 31.

## What This Changes

Nothing about our faith, our worship, or our understanding of what God accomplished through the death, burial, and resurrection of Jesus Christ depends on identifying the exact year. The cross is not more powerful if we know the calendar date. The tomb is not more empty. The text tells us what happened, and the text is what we build on.

But for those who want to see how precisely God orchestrated the fulfillment of His own pattern — the Passover lamb selected on Nisan 10, inspected for four days, killed on Nisan 14 at the appointed hour, with three days and three nights in the heart of the earth, rising on the third day — the convergence of the textual evidence and the astronomical data in a single year is remarkable.

The text tells us the sequence. The astronomy narrows the "when" to one highly probable year. Together, they point to a Wednesday in the spring of AD 31 as the day the Lamb of God was killed — at the exact hour, on the exact day, in the exact pattern that God established in Exodus 12 fifteen centuries earlier.

• • •

*The tradition is old. But the text is older. And the stars have kept God's time from the beginning.*

• • •

*All Scripture quotations are from the New American Standard Bible (NASB).*

*Astronomical data referenced from Humphreys & Waddington (Oxford, 1983/1985), Newton (1733), and Finegan's Handbook of Biblical Chronology.*

*No commentaries or denominational positions were consulted in the interpretation of the text.*

Paul & Pam • NobleMind.Study